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INTERNATIONAL EXTENDED PROJECT QUALIFICATION (9695) (Version 1.0) (Published November 2025)

Please attach the form to your candidate's work and keep it at the centre or send it to the moderator as required. The declarations should be completed as indicated.

Centre number	**Centre name**

Candidate number	**Candidate's full name**
	TANG:ZIYI

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Candidate declaration

Have you received help/information from anyone **other than** your supervisor (include specialist consultant if relevant) to produce this work?

☒ No ☐ Yes (give details below or on a separate sheet if necessary*).*

Click or tap here to enter text.

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I have read and understood the above. I confirm I produced the attached work without assistance other than that which is acceptable under the scheme of assessment.

Candidate signature

Date 2026.4.15

Supervisor declaration
I confirm the candidate's work was conducted under the conditions laid out by the specification. I have authenticated the candidate's work and am satisfied (to the best of my knowledge) that the work produced is solely that of the candidate. Artificial intelligence has not been used as the sole or primary means of marking candidates' work. Any use of AI is supplementary to the marking process and its use is continually monitored and assessed.

Supervisor signature

Date

Candidate number

Candidate's full name

TANG:ZIYI

Submission checklist

To be completed by the supervisor

International Extended Project **working** title: Click or tap here to enter text.

International Extended Project **final** title: Click or tap here to enter text.

Please note that failure to complete or submit a compulsory element may result in a mark of zero being awarded.

Select/tick	Items that **must** be included	Notes
☐	1. A signed and completed *Candidate record form, production log and assessment record*	This document. **All pages** must be completed.
☐ ☐	2. Research based written report **OR** Click here to enter text with accompanying written report	Must be **5,000 words** in length **(+/- 10%)** **OR** Photographic evidence of an artefact with an accompanying report of minimum **1,000 words** in length
☐	3. Evidence of a presentation within the *production log*	Presentation on the project process and findings/conclusions.

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The taught skills element

To be completed by the supervisor

Outline details of taught skills

Record here details of relevant skills taught in a class/group and details of relevant skills taught individually to this candidate as described in the specification eg your scheme of work. Continue on a separate sheet if necessary.

Click or tap here to enter text.

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Record of marks

To be completed by the supervisor
Marks must be awarded in accordance with the instructions and criteria in the specification.

Please circle the appropriate answer.

Has the student done the following and provided evidence of each in the submission?:

Identified and selected an area of interest.	☐Yes	☐No
Set a working title.	☐Yes	☐No
Produced a project plan.	☐Yes	☐No
Implemented the project plan and documented any changes to it.	☐Yes	☐No
Produced a project product addressing the selected final title.	☐Yes	☐No
Used a referencing method.	☐Yes	☐No
Communicated findings in the report and the presentation.	☐Yes	☐No
Created a bibliography/reference list.	☐Yes	☐No
Evaluated the strengths and weaknesses of the project.	☐Yes	☐No

If you answered "No" to any of the above, this must be reflected in the student mark.

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Candidate number	Candidate's full name
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Record of marks (continued)

To be completed by the supervisor
Marks must be awarded in accordance with the instructions and criteria in the specification.

Summary information to show how the marks have been awarded should be given in the spaces below in addition to comments in other pages of this document and any supporting information in the form of annotations on the candidate's work.

Skill area		Maximum mark	Mark awarded	Supervisor's supporting statement
AO1	Selection of topic	6	Click or tap here to enter text.	Click or tap here to enter text.
AO2	Planning, monitoring and developing	12	Click or tap here to enter text.	Click or tap here to enter text.
AO3	Demonstration of research skills	12	Click or tap here to enter text.	Click or tap here to enter text.
AO4	Analysis and application of research	24	Click or tap here to enter text.	Click or tap here to enter text.
AO5	Evaluation of product, process and self	6	Click or tap here to enter text.	Click or tap here to enter text.
Total mark		**60**	Click or tap here to enter text.	

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Record of marks (continued)

Supervisor's concluding comments

Click or tap here to enter text.

Internal moderation comments if appropriate

Click or tap here to enter text.

Supervisor declaration

I confirm that no work assessed for the award of the mark above is also to be submitted, or has been submitted, for any other accredited qualification(s).

Supervisor signature

Date

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Record of initial ideas

To be completed by the candidate
This page records initial meeting(s) with your supervisor to agree your project ideas. Additional pages can be submitted if more than one idea has been explored.

My idea(s) for topic/title:
The General Incompleteness of Formal Systems. I want to investigate whether all independence phenomena in set theory and mathematical logic (Russell's paradox, Gödel's incompleteness theorems, the independence of CH) can be unified under a single structural mechanism: self-reference combined with negation.

My preliminary research and development of my project ideas:
My interest in this area began roughly eighteen months ago, when I read Wittgenstein's Tractatus Logico-Philosophicus and from there encountered Russell's foundational programme for deriving arithmetic from logic. A specific question caught me: which is defined first, 0 or 1? I consulted specialists in mathematical logic on the. internet, who told me the answer depends on the choice of foundational model. In PA the successor function presupposes 0 as primitive; from the perspective of cardinality, existence itself appears to precede the empty. The fact that so elementary a question had no model-independent answer unsettled me.

This led me to study ZFC and the theory of cardinals and ordinals. While investigating the cardinality of the natural numbers, I encountered the Continuum Hypothesis, which struck me as a compelling open problem. At the same time, my study of Gödel's incompleteness theorems showed me that the major paradoxes and independence results in logic share a common structure: self-reference combined with negation. Russell's paradox, the Liar, Gödel's undecidable sentence, the independence of CH all seemed to me instances of the same underlying mechanism.

What follows is a summary of the notebook entries that record how my thinking developed over this period.

8 September 2025. My earliest notes on the relationship between formal systems and ontology. I wrote that there is an irreducible gap between phenomenon and noumenon, that the symbolisation of the ontological always leaves an unspeakable infinity, and that we come to know ourselves through rupture. These are not yet formal claims. They are philosophical intuitions, but they set the direction for everything that followed.

10 September 2025. I wrote: "Formalisation can never replace ontology. The existence of number is legislation, not logical deduction. The continuum, as the Real of ontology, forever resists symbolisation." This was the first time I articulated the idea that the continuum might be inherently resistant to formal determination, not merely undecided.

28 October 2025. I began writing note that attempting to state precise principles.

1. Within the symbolic logic, any concept of absence relies on the negation of presence or affirmation of a corresponding form for its existence.
2. Any form of existence is contingent upon its corresponding symbols being understood.
3. The negation of any form of existence cannot be apprehended by any form of symbol
4. Negation, as a logical operation, depends on an object of existence; we can negate what is, but not what isn't.
5. The logical form of negation is determined by its object.
6. Ontological negation resists negation

The key claims at this stage is any concept of absence depends on the negation of a corresponding presence; any form of existence depends on its corresponding symbols being understood; the negation of any form of existence cannot be apprehended by any form of symbol; negation as a logical operation depends on an existing object. I also noted that ontological negation resists negation, and that the legitimacy of a proposition depends on the unity and consistency of its logical subject.

29 October 2025. First, existence: symbolic representation refers to existing things and precedes any symbolic

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calculus; the subject's existence is not computable by any symbol. Second, negation: the subject of negation determines its form; ontological negation is not related to any predicate calculus but is a self-reflexive operation on the subject; symbols cannot represent ontological negation. Third, categories: a category is an abstraction over isomorphic representations; within the natural number category, the symbol 1 denotes the existence of an identical representation and 0 denotes its ontological negation; predicate calculus operates on all symbols except 0, and any predicate calculus satisfies two constraints: ontological symbols are indivisible and ontological negation is not computable.

1 November 2025. I reorganised the framework into a structured note with three sections: Existence, Negation, and Propositions. The section on existence states that any form of description depends on the ontological fact of its subject's existence, and that logical atomism depends on this ontological fact. The section on negation distinguishes ontological negation (negation of the subject itself) from predicate negation (negation of the copula within an assertion), and claims that ontological negation cannot itself be negated. The section on propositions defines logical propositions as depending on a subject and an assertion, and argues that for self-referential propositions such as the Liar, the subject does not exist, so the logic is invalid. This was my first attempt to dissolve the Liar paradox by denying its subject rather than its truth value.

16 December 2025. Notes influenced by Wittgenstein's Tractatus. I wrote: "Existence is a fact that is undeniable for any subject capable of recognising it. The world is the totality of existing things. The structure of the ontological constructs the representation of the world. Symbols can refer only to what exists ontologically, never to what does not exist. The boundary of symbols is the boundary of representation." This is recognisably close to Tractatus propositions 1 and 5.6, but I was arriving at them from the direction of formal decidability rather than picture theory.

30 December 2025. I wrote a set of notes attempting to formalise the common structure behind all the independence phenomena I had been studying. This was the most systematic version of the framework:

1. *general decidability principle: any act of judgement necessarily presupposes an entity that cannot itself be decided but is acknowledged as existing.*
2. *Narrow decidability principle: decidability within a formal system presupposes conditions for the system's own validity that cannot be decided within it.*
3. *Ontological constraint on judgement: any assertion that does not refer to an existing entity does not constitute an object of judgement.*
4. *Generalised tautology: the logical origin point within a formal system that the system itself cannot decide.*
5. *Generalised incompleteness principle: a formal system cannot guarantee the validity of its own generalised tautologies from within.*
6. *First generalised incompleteness theorem: any formal system capable of constructing a statement about its own totality is necessarily not closed.*
7. *Second generalised incompleteness theorem: if a formal system can refer to its own totality and possesses the capacity for negation, it necessarily constructs a self-referential negation structure, and this structure admits no final formal resolution.*

I also conceived the idea of a "decision machine": a Turing machine that would determine formal properties of axiomatic systems. These are ambitious ideas, and I am aware they may be too large for a single project, but they represent where my thinking currently stands.

My summary of the comments and advice from my supervisor (and specialist consultant if applicable):

My supervisor, a mathematics teacher, considered the scope of "generalised incompleteness" too ambitious for an EPQ. He noted that constructing an entirely new meta-logical framework would be difficult to assess within the EPQ marking criteria and advised me to narrow the project to a more specific, analytical question. He recommended focusing on a concrete problem within set theory rather than attempting a grand unification, and suggested that an analytical investigation examining existing results and their implications would be more appropriate than a purely original theoretical construction.

Modifications I have made as a result of my discussion with my supervisor (and specialist consultant if applicable):

I have decided to abandon the "generalised incompleteness" framework as the explicit subject of the project. Shortly after the meeting, I discovered through further reading that the unifying structure I had identified is

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essentially equivalent to Lawvere's fixed point theorem, a result already established in category theory. This confirmed my supervisor's judgement: the direction was not viable for an EPQ. The part that was original could not be made rigorous enough within the constraints; the part that could be made rigorous was not original.

The question then was what to replace it with. My supervisor had advised me to focus on a concrete problem in set theory. The Continuum Hypothesis was the natural choice. I had already encountered CH months earlier, while studying cardinals and ordinals. At the time I was trying to understand how infinite cardinalities behave, and CH appeared as the first and most striking open question: is there a cardinality between the naturals and the reals? Its independence from ZFC, proved by Gödel and Cohen, immediately struck me as potentially connected to the incompleteness phenomena I had been studying. Both seemed to be cases where a formal system fails to determine the truth of a statement that ought to have a definite answer. That intuition, that CH's independence might share a structural root with Gödel's theorems, was what had drawn me to it in the first place. It was not a new interest but the oldest one I had in set theory.

My supervisor's advice was to take a specific, assessable problem and analyse it thoroughly rather than attempt a grand unification. CH was the problem I knew best, and the question it raised, whether new axioms genuinely resolve it or merely select a universe, was analytical rather than speculative. I could pursue it with the tools of mathematical logic rather than with the untested framework I had just abandoned.

I plan to investigate whether and how new axioms can resolve CH within or beyond ZFC. The philosophical questions that motivated the earlier framework have not disappeared, but they are not the reason I chose this particular problem.

My proposed title, aims and objectives:

Aim: To investigate why the Continuum Hypothesis has no answer within ZFC, and to analyse the extent to which proposed new axioms can genuinely resolve it.

Objectives:

1. Analyse the technical mechanisms behind CH's independence from ZFC (Gödel's constructible universe and Cohen's forcing).
2. Evaluate the major proposed resolutions: V = L, large cardinal axioms, forcing axioms, and Woodin's Ultimate L programme.
3. Investigate whether the boundary between resolvable and irresolvable propositions can itself be determined.

Date 2026.1.11

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Candidate number	Candidate's full name
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Part A: Candidate proposal

To be completed by the candidate

Working title of my International Extended Project.
Present the topic to be researched in the form of a short statement/question/hypothesis with clear focus.
Beyond ZFC: New Axioms and the Continuum Hypothesis. To what extent is the introduction of new axioms a justified and necessary approach to the Continuum Hypothesis?

My initial resources will be
Book and Paper:
Gödel, K. Über formal unentscheidbare Sätze" (1931)
Gödel, K. Consistency of CH via the constructible universe L (1938, 1939, 1940)
Cohen, P. The independence of CH via forcing (1963, 1966)
Gödel, K. "What is Cantor's Continuum Problem?" (1947)
Lévy, A. and Solovay, R. Measurable cardinals and the continuum hypothesis (1967)

YouTube video:
The Continuum Hypothesis and the search for Mathematical Infinity, W. Hugh Woodin(https://www.youtube.com/watch?v=nVF4N1Ix5WI&t=277s)
Set Theory and the Philosophy of Set Theory,Joel David Hamkins (https://www.youtube.com/watch?v=v_tPUH5NtwI)

AI:
Claude opus4.6(asking for explain detial)

The courses of study or area(s) of personal interest to which the topic relates
Mathematics, Further Mathematics, Mathematical Logic, Philosophy of Mathematics

My project aims and objectives
The project aims to determine whether the Continuum Hypothesis can be genuinely resolved by any axiom appended to ZFC, and if not, to characterise why not. The objectives are:
(1) to analyse the independence of CH through the constructions of Gödel and Cohen.
(2) to evaluate V = L, large cardinal axioms, forcing axioms, and Ultimate L as candidate resolutions.
(3) to develop a precise criterion distinguishing genuine resolution from universe-selection.
(4) to investigate whether such a criterion can be uniformly applied, that is, whether the classification of propositions into resolvable and irresolvable is itself decidable.

Provide details of the courses that you are currently studying

Qualification type	Awarding body and subject
eg A-level, IB, Modern Apprenticeship, BTEC	eg AQA Mathematics, OCR Computing, WJEC English
A Level	CAIE 9709 Mathematics
A Level	CAIE 9231 Further Mathematics
A Level	CAIE 9618 Computer Science

Qualification type	Awarding body and subject
A Level	CAIE 9702 Physics
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Candidate declaration

I certify that I have read and understood OxfordAQA Regulations relating to unfair practice as set out in the notice to candidates above.

Candidate signature

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Part B: Supervisor's comments on candidate proposal

To be completed by the supervisor
Please comment below on the validity and feasibility of the candidate proposal (Part A) as an International Extended Project.

	Supervisor's comments
Indicate the relation to, and development/extension outside of, the main course(s) of study or interest.	Click or tap here to enter text.
Comment on the suitability of the proposed initial sources and research base.	Click or tap here to enter text.
Confirm that the project is feasible in the proposed timescale and/or indicate any potential difficulties that may prevent the candidate from meeting the assessment objectives.	Click or tap here to enter text.
Comment on the suitability of the proposed title, aims and objectives.	Click or tap here to enter text.

Supervisor signature

Date

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Candidate number

Candidate's full name

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Part C: Centre coordinator's approval of candidate proposal

Supervisor's name

Click or tap here to enter text.

To be completed by the centre coordinator
If you are acting as both the centre coordinator and the supervisor, please seek counter signature from a senior colleague.

Centre coordinator's comments on the feasibility and acceptability of the proposal (parts A and B) as an International Extended Project Qualification.

Click or tap here to enter text.

☐ Approved

☐ Approved subject to the implementation of the centre coordinator's recommendations

☐ Resubmission required

Centre coordinator's name

Click or tap here to enter text.

Centre coordinator signature

Date

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Planning review

To be completed by the candidate

This page records your outline plan once your proposal has been approved.

My next steps in planning, researching and deadlines that I will set myself. What I intend to do, by when, what resources I will use and how I will implement the recommendations of the centre coordinator (where appropriate).

January 2026: Complete reading of primary sources. Gödel (1931, 1938 to 1940, 1947), Cohen (1963, 1966), Skolem (1922). Begin mapping the logical structure of the essay.

February 2026: Complete reading of secondary sources. Maddy (1988), Woodin (2001), Scott (1961), Lévy and Solovay (1967). Complete a first draft covering the four main sections: CH and ZFC; V = L; large cardinals and forcing axioms; a concluding analysis.

March 2026: Submit draft to supervisor for review. Revise based on feedback. Develop any original arguments that emerge during writing.

April 2026: Final revision, proofreading, plagiarism check, and submission.

The reasons for my decisions.

I chose to read primary sources rather than relying solely on textbook expositions. My research question asks whether new axioms genuinely resolve CH or merely select a universe. To answer this, I need to understand not just the results but the constructions that produce them: the specific mechanisms by which Gödel's constructible universe forces CH to hold and Cohen's generic extensions force it to fail. A textbook summary tells you that V = L implies CH; Gödel's own construction shows you why, and the difference matters for my argument.

The essay's structure follows my own learning trajectory. I first encountered CH through the study of cardinals, then believed V = L offered a satisfying resolution, then realised that the choice between V = L and its alternatives depends on one's philosophical stance toward the set-theoretic universe. This intellectual path became the structure: Section I poses the question, Section II examines contraction (V = L), Section III examines expansion (large cardinals, forcing axioms), and Section IV confronts the underlying issue.

The fourth objective, on the decidability of the resolvable/irresolvable boundary, comes from my earlier "generalised incompleteness" work, where I had conceived a "decision machine" for formal systems. That framework was abandoned, but the idea survived. I do not yet know whether it is tractable within this project, but I want to keep it as a possibility.

My summary of the comments and advice from my supervisor (and specialist consultant if applicable).

My supervisor reviewed the draft outline and reading list. He approved the analytical direction and encouraged me to go beyond a historical survey. The essay should not merely recount what each mathematician proved but should develop a coherent argument evaluating whether these results genuinely resolve CH. He also noted a risk in my reading plan: primary sources in mathematical logic are dense, and attempting to master every technical detail might consume time better spent on the analytical argument. He advised me to distinguish between understanding a construction well enough to evaluate its philosophical implications and understanding it well enough to reproduce every proof. The former is necessary; the latter is not.

Modifications I have made as a result of my discussion with my supervisor (and specialist consultant if applicable).

I adjusted my reading strategy in light of my supervisor's warning about primary sources. Rather than working through every proof in Gödel's and Cohen's original papers line by line, I would read the constructions closely enough to understand the model-theoretic mechanism and then move on to the analytical consequences. In practice this meant spending less than a week on each primary source. When a proof or construction was too dense to follow in the original, I used Wikipedia and other internet resources to get a rapid overview of the key ideas and then returned to the source with that understanding in place. I also decided to begin drafting the essay outline while still reading, rather than waiting until all reading was complete, so that the structure of the argument

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could take shape alongside the technical understanding rather than after it.

A separate change came from a philosophical essay I was writing concurrently on theology. That essay used a reductio ad absurdum as its governing structure, and I realised the same approach suited my EPQ better than the expository format I had been planning. I restructured the argument so that instead of presenting axiom candidates in sequence and then evaluating them, the essay would assume a genuine resolution exists and show that the assumption destroys itself. This tightened the argumentative chain and substantially reduced the essay's length.

Please append any relevant planning documentation or tools that you have used.

Date 16 January 2026

Candidate number	**Candidate's full name**
	Click or tap here to enter text.

Mid-project review

To be completed by the candidate

This page records your outline plan when you have completed your research.

Is my project following my proposed plan? How has my plan developed since my project proposal?

The project is not following plan. I have had to make a significant change of direction.

The first draft followed the original plan faithfully across Sections I through III. Section I introduced CH and its independence from ZFC. Section II analysed V = L as a proposed resolution. Section III examined large cardinal axioms, forcing axioms, and Woodin's Ultimate L programme. These sections worked as intended.

The problem was Section IV. What was meant to be a concluding analysis grew into something far larger. During drafting I developed a complete theoretical framework that I called "horizon theory." The central idea was that every formal system has two cognitive boundaries. The first is an inner horizon, the point at which self-reference breaks down. A system can verify the consistency of systems weaker than itself, but it cannot verify its own consistency, and this boundary where self-verification fails is the inner horizon. The second is an outer horizon, the highest ordinal the system can reach in its proofs. For Peano Arithmetic, for example, this is the ordinal epsilon-nought. Everything the system can decide lies between these two boundaries, and everything outside them is undecidable.

I used this framework to classify independence phenomena into two types. Inner-type independence is driven by self-reference. The consistency of PA is independent of PA because PA hits the inner wall and cannot verify itself. Outer-type independence is driven by ordinal strength. Goodstein's theorem is independent of PA because proving it requires ordinal epsilon-nought, which PA cannot reach. Both types coexist in the same system, and I argued this was not coincidence but a structural necessity.

I then proposed three claims. A Horizon Hypothesis, stating that all decidable statements of a formal system lie strictly within the inner and outer horizons and that neither horizon can be crossed. A Conservation Conjecture, stating that the gap between the two horizons is invariant under consistent axiomatic extensions, so that adding axioms moves both boundaries but the distance between them is preserved. And a Forcing Duality Conjecture, stating that every model-theoretic operation on the outer horizon has a corresponding proof-theoretic operation on the inner horizon and vice versa, which would provide a bridge between set theory and proof theory.

Looking at this now, it reads like an independent research programme, not like the conclusion of an EPQ essay. The conjectures are unproven and I am not sure they are provable within this project. The notation looks impressive on the page but I am not confident it does real mathematical work.

I showed this draft to my supervisor and his feedback confirmed what I was already suspecting. At the same time I have been reading more widely in model theory, and I have come across the concept of absoluteness, the property that a formula's truth value is the same in every transitive model of a theory. I am realising that absoluteness already does, with full mathematical rigour, what my horizon theory was trying to do. The boundary I was drawing

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between inner-type and outer-type independence is essentially the boundary between absolute and non-absolute formulas. Absoluteness has precise definitions, well-understood properties, and deep connections to the Shoenfield theorem and the Lévy hierarchy. My horizon notation has only suggestive appeal.

I have decided to abandon the horizon theory and restructure the essay around absoluteness as a criterion for genuine resolution. I am also thinking about whether the "decision machine" idea from my earlier project might be recoverable here. If absoluteness is the right criterion, can the question "is this sentence absolute?" be decided by an algorithm? I do not know the answer yet, but the halting problem seems like it might be relevant.

My summary of the comments and advice from my supervisor (and specialist consultant if applicable).

My supervisor reviewed the first draft and made two observations. The horizon theory was too speculative and too original for an EPQ. It resembled an independent research programme rather than an analytical essay. He recommended replacing it with something grounded in established mathematical concepts, where the originality would lie in applying established tools to a specific question rather than inventing new tools. He also reiterated a principle from our earlier discussions, "be original in analysis, not in axiomatics," which captured exactly what I had been struggling to articulate.

Modifications I have made as a result of my discussion with my supervisor at this stage (and specialist consultant if applicable).

I am abandoning the horizon theory and restructuring the essay around the absoluteness criterion.

Section I will define absoluteness and use the consistency of PA as a precedent, since Con(PA) is a truth that is absolute across all transitive models and therefore counts as a genuine resolution. Section II will analyse V = L as ontological pruning, a form of universe-selection that works by restricting to definable sets. Section III will analyse large cardinal axioms, focusing on the Lévy-Solovay orthogonality result, and forcing axioms such as Martin's Maximum, and will then establish that the cardinality of the power set of omega is not absolute. Section IV will develop the philosophical argument that specifying the true universe V is circular, because singling out V requires specifying which subsets of omega exist, which is precisely the question CH asks. Section V will attempt to prove that absoluteness cannot be uniformly decided. I believe a reduction to the halting problem should work, but I have not completed the construction yet. If it does not work I will scale this section back.

The essay is being retitled. I am considering "On the Indeterminacy of the Continuum Hypothesis and the Incomputability of Indeterminacy".

My final title, aims and objectives.

Title: To What Extent Is the Resolution of the Continuum Hypothesis a Matter of Mathematical Discovery Rather Than Ontological Selection?

Research question: Can any axiom appended to ZFC genuinely resolve whether the Continuum has a definite cardinality, and can this question itself be decided?

Aims and objectives:

1. Define a precise criterion for "genuine resolution" using absoluteness across transitive models of ZFC.
2. Demonstrate that V = L, large cardinal axioms, forcing axioms, and Ultimate L all fail this criterion.
3. Establish the non-absoluteness of the cardinality of the power set of omega and argue that CH is not merely undecidable but indeterminate.
4. Prove that the absoluteness criterion cannot be uniformly decided (by reduction to the halting problem.

My planned next steps to achieve my project's aims and objectives.

Complete the rewrite of Sections I through V. The first four sections should come together fairly quickly since the analytical structure is now clear. Section V depends on whether the halting-problem reduction actually goes through. Proofread for mathematical accuracy and argumentative coherence. Run plagiarism check. Prepare for the oral presentation.

Please append any relevant planning documentation or tools that you have used.

Date

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Project product review

To be completed by the candidate

This page records the near completion of your project product. Outline the successes, failures, additions and/or changes you made as you followed the plan in your mid-project review.

Did my project follow my revised plan from the mid-project review?

Yes. The rewrite was completed. The final essay comprises five sections structured around the absoluteness criterion. Sections I through IV analyse CH through the lens of absoluteness, evaluating each major proposed resolution and showing that none achieves invariance across transitive models. Section V proves that the absoluteness criterion itself cannot be uniformly decided, by constructing a sentence whose absoluteness is equivalent to the halting of an arbitrary Turing machine.

The halting-problem reduction did work. For an arbitrary Turing machine M, the sentence "M halts or CH" is absolute if and only if M halts. If M halts, the first disjunct is a true arithmetic statement, absolute everywhere. If M does not halt, the sentence reduces to CH, which is not absolute. A decision procedure for absoluteness would therefore solve the halting problem, which is impossible. The construction came together during the writing of Sections III and IV, when I saw how the halting problem could supply the method while absoluteness supplied the target.

The one element that exceeded the mid-project plan was Section IV. I had conceived it as a brief philosophical bridge, but during writing a sustained argument emerged that I had not anticipated. The key insight was a circularity: any attempt to single out the "true" universe V from among transitive models sharing the same membership relation, the same natural numbers, and the same large cardinals must appeal to which subsets of omega the universe contains, but specifying this is precisely the question CH asks. This circularity is not epistemic but structural. It transformed the essay's conclusion from "CH cannot be resolved" to "CH cannot be resolved, and this is because of what the power set of omega is."

I recognise now that this circularity argument is the mathematical formulation of a philosophical intuition I had recorded in my notebook over a year ago: that formalisation encounters an irreducible limit when confronting certain totalities. The conviction that drove the ontological notebook, the generalised incompleteness framework, and the horizon theory, each of which failed to express it rigorously, finally found its proper language in the standard vocabulary of model theory.

My summary of the comments and advice from my supervisor at this final stage (and specialist consultant if applicable).

My supervisor reviewed the final draft and approved the restructured argument. He noted that the absoluteness criterion provided a much clearer analytical framework than the earlier horizon theory, and that Section V's reduction to the halting problem constituted a genuine and well-grounded original contribution. He suggested minor clarifications in the footnotes regarding the meta-theoretic assumptions, specifically the presupposition that transitive models of ZFC exist, which cannot be established within ZFC itself. He also recommended changing the title. The working title, "On the Indeterminacy of the Continuum Hypothesis and the Incomputability of Indeterminacy," was too general: it described the content but did not frame a question. He suggested reformulating it as a question that captured the central tension of the essay. He confirmed the essay was ready for submission after these revisions.

Modifications I have made as a result of discussion with my supervisor (and specialist consultant if applicable) at this final stage.

Do I need to do anything else to complete my project?

I added clarifying footnotes on the meta-theoretic assumptions. The footnotes now state explicitly that the argument presupposes the existence of transitive models of ZFC as a working hypothesis, a hypothesis that holds if there exists an inaccessible cardinal but that cannot be proved within ZFC itself. I also changed the title to: "To What Extent Is the Resolution of the Continuum Hypothesis a Matter of Mathematical Discovery Rather Than Ontological

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Selection?" This formulation captures the essay's core argument more precisely than the previous title, which merely named the two results without framing the analytical question. The essay has passed plagiarism checking. The remaining task is to prepare and deliver the oral presentation.

Please append any relevant planning documentation or tools that you have used.

Date 2026.4.3

Candidate number	Candidate's full name
Click or tap here to enter text.	TANG:ZIYI

Presentation record part A

To be completed by the candidate
This page records your presentation and its preparation.

Planned format of my 10 minute presentation (eg audience, use of visual aids, slides, use of notes, etc).

The audience will include my supervisor, who teaches mathematics, and one additional assessor.

I prepared a slide deck using LaTeX Beamer, which allowed me to typeset the mathematical notation cleanly. The presentation consists of six slides. I chose LaTeX over PowerPoint because the essay contains a high density of formal symbols, including set-theoretic notation, cardinal arithmetic, and the sentence construction for the halting-problem reduction, and these would have been difficult to render legibly in a standard presentation tool. The visual style of the slides was modelled on Woodin's lecture slides, which I had been reading throughout the project. His slides present dense mathematical content with minimal decoration and let the formal statements speak for themselves, and I wanted mine to have the same quality.

Planned content of my presentation.

- Project process:

The six slides are structured as follows. Slide 1 is the title slide. I state my name and the project title, then present the two main claims upfront. The first is that the Continuum Hypothesis does not have a definite answer. The second is that the criterion I use to show this, absoluteness, is itself incomputable. Slide 2 introduces the problem. I present Cantor's result that the reals are strictly larger than the naturals, state CH as the claim that there is nothing in between, and summarise the independence results of Gödel in 1938 and Cohen in 1963. I then pose the research question, whether any axiom can genuinely resolve CH or whether every resolution merely selects a universe. Slide 3 defines the criterion and the key facts. A sentence is absolute if every transitive model of set theory gives it the same truth value. I explain the asymmetry between the natural numbers, which are absolute across transitive models, and the power set of the natural numbers, which is not. Different models contain different subsets. CH asks about the cardinality of the power set, so CH is not absolute. Slide 4 examines three strategies and their shared failure. V = L restricts to definable sets and discards models with undefinable subsets. Large cardinals, by the Lévy-Solovay theorem, are orthogonal to CH. Forcing axioms such as Martin's Maximum stipulate which generic objects the universe contains. All three fail the absoluteness criterion. Slide 5 presents the incomputability of absoluteness. For any Turing machine M, I define the sentence "M halts or CH." If M halts, the sentence is absolute because it reduces to a true arithmetic statement that is invariant across models. If M does not halt, the sentence reduces to CH, which is not absolute. A decision procedure for absoluteness would therefore solve the halting problem, which is impossible.

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- Project findings and conclusions:

Slide 6 draws conclusions. CH does not express a determinate proposition. The variability of its truth value reflects the fact that the power set of the natural numbers has no stable referent. The boundary between what set theory can determine and what it cannot is itself beyond algorithmic reach.

Modifications I have made as a result of rehearsal and/or discussion with my supervisor (and specialist consultant if applicable):

After a practice run with my supervisor, I made two adjustments. I was spending too much time on the historical background of CH in Slide 2, leaving insufficient time for the theorem in Slide 5. I cut the history down to the minimum needed and moved the remaining time to the incomputability argument. My supervisor also suggested that I state the two main results at the very beginning rather than saving them for the end, so the audience knows from the start what the presentation is building toward. Both adjustments were incorporated into the final version of the slides.

Please append any relevant material you have prepared to support your presentation.

Date 2026.4.12

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Candidate number	Candidate's full name
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Presentation record part B

To be completed by the supervisor
Record and comment below on the delivery of the presentation.

	Supervisor's record/comments
The nature of the audience should be a minimum of two adults including supervisor (include numbers of staff, students and others present).	Click or tap here to enter text.
The nature of the presentation (include use of notes, use of display items, and use of presentation software).	Click or tap here to enter text.
Comment on the content and delivery of the presentation (for example clarity of ideas, structure of presentation, pace, engagement with audience).	Click or tap here to enter text.

Please append any relevant planning documentation or tools that you have used.
Do not include any video evidence or observer notes as these are not required

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Presentation record part B

To be completed by the supervisor
Record and comment below on the delivery of the presentation.

Comment on the response of the candidate to questions that demonstrated understanding and grasp of the project and/or its production. **Give examples of five questions asked and answers given.**	1. Click or tap here to enter text.
	2. Click or tap here to enter text.
	3. Click or tap here to enter text.
	4. Click or tap here to enter text.
	5. Click or tap here to enter text.

Supervisor signature

Date

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Summary and reflection

To be completed by the candidate
This page records your summary, reflection and evaluation of your planning, the final outcome and your personal learning. It should be written after you have completed your report and given your presentation.

Some questions you may wish to answer in this section include, what have I learned from completing this project? What new knowledge or expertise have I enjoyed or found valuable? What are the strengths and weaknesses of my project (including planning and organisation)? What skills have I improved? What changes would I make if I undertook such work again? What advice would I give to others undertaking such a project?

The project I finished is not the project I set out to write. The original plan was to investigate generalised incompleteness, a framework unifying all independence phenomena in logic. That was abandoned before the project formally began. The replacement plan was to investigate CH through the lens of new axioms, with an original concluding analysis. That plan held through Sections I to III, but the concluding analysis grew into an independent theoretical framework (horizon theory) that was too speculative to survive scrutiny. The final essay emerged only after this second failure, when I restructured everything around the absoluteness criterion and discovered that the "decision machine" idea from the earliest stage of my research could be recovered as a theorem.

What I learned about research is that ideas do not arrive in order. The most important result in my essay (Section V) originated from a question I asked months before I had any of the tools to answer it. The question survived the destruction of two separate theoretical frameworks and only found a home when I stumbled upon the right mathematical concept. I could not have planned this. It happened because I kept the question alive while letting go of the contexts that had produced it.

What I learned about writing is that structure matters more than I expected. The difference between my horizon theory draft and the final essay is not that one contains better ideas than the other. The ideas are recognisably related. The difference is that the final essay is organised around a single, precise criterion (absoluteness) that every section tests and extends, whereas the horizon theory draft was organised around a notation that looked impressive but did no real work. My supervisor's phrase, "be original in analysis, not in axiomatics," became a principle I will carry into future work.

What I learned about time management is that I underestimated how long primary sources take to read. Cohen's forcing monograph consumed two weeks that I had allocated one week for, and my supervisor's early warning about this proved correct. I compensated by shifting my focus from technical mastery to model-theoretic consequences, which preserved my timeline but required a deliberate decision about what to sacrifice. In future projects I would build more buffer time into reading schedules for technically demanding sources.

What I learned about presentation is that a ten-minute talk on a technical subject forces a different kind of understanding. Writing the essay, I could develop an argument across pages and let the reader follow at their own pace. Standing in front of an audience, I had to know which steps could be compressed and which had to be stated in full. The practice run showed me that I was treating Slide 2 as a miniature lecture on the history of CH, when all the audience needed was the independence result and the question it raises. Cutting that material and reallocating time to the incomputability argument in Slide 5 improved the presentation not just in pacing but in clarity, because it forced me to identify what the talk was actually about. My supervisor's suggestion to state both main results on the first slide, rather than building toward them, changed the entire dynamic. The audience listened differently when they knew from the beginning what I was going to show them.

The project is not finished in my mind. The EPQ established that absoluteness is incomputable by a single reduction to the halting problem, but this leaves open the question of exactly how incomputable it is. I am now planning a follow-up project in two parts. The first is a machine-verified formalization in Isabelle/ZF. I intend to define Turing machines, oracle Turing machines, and the Turing jump within ZF set theory, then formally prove that absoluteness determination is not computable relative to any finite iterate of the Turing jump. This would show that absoluteness lies above every level of the arithmetic hierarchy, not just above the first level as my EPQ argument demonstrates. The second part is a technical paper investigating the precise descriptive complexity of the

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absoluteness problem. I believe it is Pi-one-one complete in the framework of omega-models of second-order arithmetic. The upper bound argument uses a monotonicity property of Pi-one-one sentences across omega-models, and the lower bound proceeds by reducing the well-foundedness problem for trees to absoluteness, using the fact that RCA-nought does not prove ATR-nought and therefore its omega-models can omit paths through non-well-founded trees. This is speculative and may not work, but the Isabelle formalization is concrete and I have already drafted a detailed plan for it.

If I were advising a student considering a project at this level of mathematical abstraction, I would say three things. First, read primary sources, not just textbooks; the constructions contain information that summaries flatten. Second, be prepared to abandon frameworks that are not working, even if you are attached to them; the quality of the final product depends on your willingness to destroy intermediate products. Third, keep a record of your ideas even when they seem premature or disconnected; you may find a use for them much later, in a context you cannot currently foresee.

2026.4.15

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